

Problem 73. A triangle ABC with $AB < AC$ is inscribed in a circle ω . Circles γ_1 and γ_2 touch the lines AB and AC , and their centres lie on the circumference of ω . Prove that C lies on a common external tangent to γ_1 and γ_2 .

Problem 74. Find all positive integers $n \geq 2$ such that there exists a permutation $a_1, a_2, a_3, \dots, a_{2n}$ of the numbers $1, 2, 3, \dots, 2n$ satisfying

$$a_1a_2 + a_3a_4 + \dots + a_{2n-3}a_{2n-2} = a_{2n-1}a_{2n}.$$

Problem 75. A calculator can square a number or add 1 to it. It cannot add 1 two times in a row. After several operations it transformed a number x into a number $S > x^n + 1$ where x, n, S are positive integers. Prove that $S \geq x^n + x - 1$.

Problem 76. Determine if there exist positive integers $a_1, a_2, \dots, a_{10}, b_1, b_2, \dots, b_{10}$ satisfying the following property: for each non-empty subset S of $\{1, 2, \dots, 10\}$ the sum $\sum_{i \in S} a_i$ divides the sum $12 + \sum_{i \in S} b_i$.